

Access Modifiers

1. Create a class with PRIVATE fields, private method and a main method. Print the fields in main method. Call the private method in main method.

Create a sub class and try to access the private fields and methods from sub class.

Answer:

Parent.java

```
class Parent {  
  
    private int number = 100;  
  
    private void display() {  
        System.out.println("Private Method");  
    }  
  
    public static void main(String[] args) {  
        Parent obj = new Parent();  
  
        // Accessing private field  
        System.out.println("Number = " + obj.number);  
  
        // Accessing private method  
        obj.display();  
    }  
}
```

Child.java

```
class Child extends Parent {  
  
    public static void main(String[] args) {  
        Child obj = new Child();  
  
        // Not Allowed  
        // System.out.println(obj.number);  
        // obj.display();  
  
        System.out.println("Cannot access private members.");  
    }  
}
```

Output

Number = 100

Private Method

Cannot access private members.

2. Create a class with DEFAULT fields and methods. Access these fields and methods from any other class in the same package ?

Answer:

DefaultClass.java

```
class DefaultClass {  
  
    int number = 200;        // Default field  
  
    void display() {         // Default method  
        System.out.println("Default Method");  
    }  
}
```

TestDefault.java

```
class TestDefault {  
  
    public static void main(String[] args) {  
  
        DefaultClass obj = new DefaultClass();  
  
        System.out.println(obj.number);  
        obj.display();  
    }  
}
```

Output

```
200  
Default Method
```

3. Create a class with PROTECTED fields and methods. Access these fields and methods from any other class in the same package.

Also, Access the PROTECTED fields and methods from child class located in a different package

Access the PROTECTED fields and methods from any class in different package

Answer :

Package 1

package1/ProtectedClass.java

```
package package1;
```

```
public class ProtectedClass {
```

```
protected int number = 300;

protected void display() {
    System.out.println("Protected Method");
}
}
```

package1/TestProtected.java

```
package package1;

public class TestProtected {

    public static void main(String[] args) {

        ProtectedClass obj = new ProtectedClass();

        System.out.println(obj.number);
        obj.display();
    }
}
```

Output

```
300
Protected Method
```

Child class in another package

package2/Child.java

```
package package2;

import package1.ProtectedClass;

public class Child extends ProtectedClass {

    public static void main(String[] args) {

        Child obj = new Child();

        System.out.println(obj.number);
        obj.display();
    }
}
```

Output

300

Protected Method

Another class in different package

package2/Test.java

```
package package2;
```

```
import package1.ProtectedClass;
```

```
public class Test {
```

```
    public static void main(String[] args) {
```

```
        ProtectedClass obj = new ProtectedClass();
```

```
        // Not Allowed
```

```
        // System.out.println(obj.number);
```

```
        // obj.display();
```

```
        System.out.println("Cannot access protected members  
using object in different package.");
```

```
    }
```

```
}
```

Output

Cannot access protected members using object in different package

4. Create a class with PUBLIC fields and methods.

Access the public methods and fields from any class in the same package or different package.

Answer :

Package 1

package1/PublicClass.java

```
package package1;
```

```
public class PublicClass {
```

```
    public int number = 400;
```

```

        public void display() {
            System.out.println("Public Method");
        }
    }
package1/TestPublic.java
package package1;

public class TestPublic {

    public static void main(String[] args) {

        PublicClass obj = new PublicClass();

        System.out.println(obj.number);
        obj.display();
    }
}

```

Different Package package2/Test.java

```

package package2;

import package1.PublicClass;

public class Test {

    public static void main(String[] args) {

        PublicClass obj = new PublicClass();

        System.out.println(obj.number);
        obj.display();
    }
}

```

Output

```

400
Public Method

```